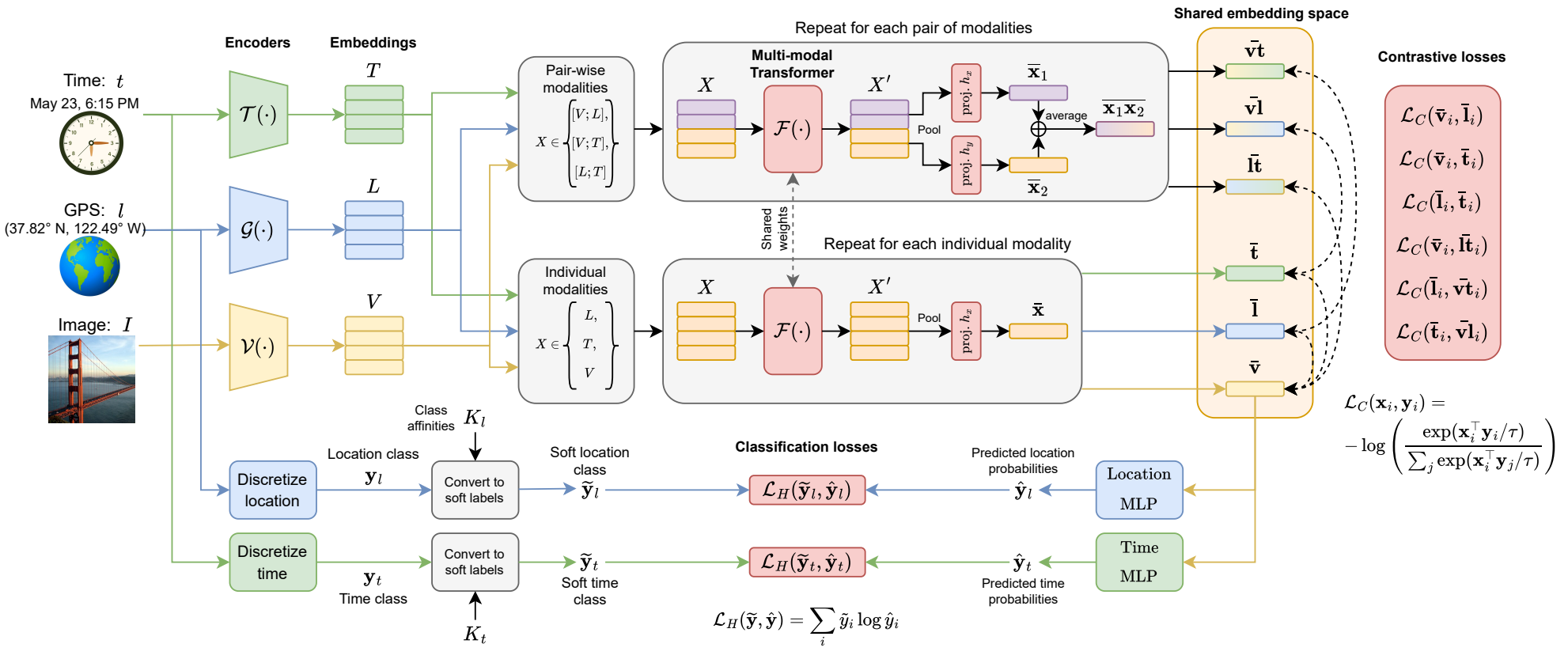
Time: t
May 23, 6:15 PM
GPS: l
(37.82° N, 122.49° W)
Image: I
Encoders
Embeddings
$\mathcal{T}(\cdot)$
$\mathcal{G}(\cdot)$
$\mathcal{V}(\cdot)$
T
L
V
Pair-wise modalities
$X \in \{[V;L], [V;T], [L;T]\}$
Individual modalities
$X \in \{L, T, V\}$
Repeat for each pair of modalities
Multi-modal Transformer
X
$\mathcal{F}(\cdot)$
X'
Pool
proj. h_x
proj. h_y
$\bar{\mathbf{x}}_1$
$\bar{\mathbf{x}}_2$
average
$\overline{\mathbf{x}_1\mathbf{x}_2}$
Shared weights
Repeat for each individual modality
$\bar{\mathbf{x}}$
Shared embedding space
$\bar{\mathbf{v}}\bar{\mathbf{t}}$
$\bar{\mathbf{v}}\bar{\mathbf{l}}$
$\bar{\mathbf{l}}\bar{\mathbf{t}}$
$\bar{\mathbf{t}}$
$\bar{\mathbf{l}}$
$\bar{\mathbf{v}}$
Contrastive losses
$\mathcal{L}_C(\bar{\mathbf{v}}_i, \bar{\mathbf{l}}_i)$
$\mathcal{L}_C(\bar{\mathbf{v}}_i, \bar{\mathbf{t}}_i)$
$\mathcal{L}_C(\bar{\mathbf{l}}_i, \bar{\mathbf{t}}_i)$
$\mathcal{L}_C(\bar{\mathbf{v}}_i, \overline{\mathbf{l}\mathbf{t}}_i)$
$\mathcal{L}_C(\bar{\mathbf{l}}_i, \overline{\mathbf{v}\mathbf{t}}_i)$
$\mathcal{L}_C(\bar{\mathbf{t}}_i, \overline{\mathbf{v}\mathbf{l}}_i)$
$\mathcal{L}_C(\mathbf{x}_i, \mathbf{y}_i) = -\log\left(\frac{\exp(\mathbf{x}_i^\top \mathbf{y}_i/\tau)}{\sum_j \exp(\mathbf{x}_i^\top \mathbf{y}_j/\tau)}\right)$
Class affinities K_l
Location class $\mathbf{y}_l$
Discretize location
Convert to soft labels
Soft location class $\tilde{\mathbf{y}}_l$
Classification losses
$\mathcal{L}_H(\tilde{\mathbf{y}}_l, \hat{\mathbf{y}}_l)$
Predicted location probabilities $\hat{\mathbf{y}}_l$
Location MLP
Discretize time
$\mathbf{y}_t$ Time class
Convert to soft labels
K_t
Soft time class $\tilde{\mathbf{y}}_t$
$\mathcal{L}_H(\tilde{\mathbf{y}}_t, \hat{\mathbf{y}}_t)$
Predicted time probabilities $\hat{\mathbf{y}}_t$
Time MLP
$\mathcal{L}_H(\tilde{\mathbf{y}}, \hat{\mathbf{y}}) = \sum_i \tilde{y}_i \log \hat{y}_i$